

GAYRARD TRISTAN

Master's Research Intern

EDUCATION

Master's degree

Ocean, Atmospheric and Climate Sciences

University of Toulouse, France

2025 - now

Bachelor degree

Applied Physics - Specialized in Materials and Physico-Chemical Controls

University of Toulouse, France

2022 - 2025

PROFESSIONAL EXPERIENCES

Internship - LAERO

Quantification and identification of combustion sources by low-cost sensors

Toulouse, France

2026

- Operated "MICROMEGAS", a lab-developed instrument.
- Conducted a measurement campaign at the Lannemezan Atmospheric Research Centre for calibration data acquisition.
- Determined tracer-tracer signatures for combustion source identification.
- Developed custom software using Python & PyQt6 for instrument calibration.

Internship - University of Wyoming

Drone-Based Nitrous Oxide & Methane Leak Detection

Laramie, WY, USA

April - June 2024

- Calibrated and operated drone-mounted spectrometers.
- Executed a field measurement campaign on an active mining/drilling site.
- Designed graphical user interfaces using Python Tkinter to process sensor calibration and data treatment.

Apprenticeship - EvO Metrology

Accreditation Substitute & continuous lab improvements

Toulouse, France

2024 - 2025

- Ensured compliance with metrology standards and accreditation procedures.
- Developed software to automate lab processes.

CONTACT

Email: tristan.gayrard@utoulouse.fr

Phone: +33 7 83 32 20 87

Location: Toulouse, France

LinkedIn



GitHub



Portfolio



HARD SKILLS

- Developing Environmental Assessments
- Implementing Innovative Methodologies
- Metrology & quality procedures
- Measurement campaign execution
- Digital tool development
- Critical analysis & argumentation
- Data treatment, analysis & synthesis
- Project teamwork & management

PERSONAL PROJECTS

- Planetary Nebula Analysis
Graphical UI with PyQt6 and Python for nebula class determination (Guaradian method)
- Mini-Weather Station
Arduino and BME680 powered, OLED display, and custom-built rain gauge

LANGUAGES

French

mother tongue

English - C1

everyday and technical language

INTERESTS & OTHER

- Electronics, Fishing, Mountain Biking
- Valid driving licence, First Aid diploma